

Statement of Research Interests

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My research focuses on how environmental exposures and policy changes affect health outcomes and economic wellbeing. An existing challenge in this area is that pollution, health behaviors, and policy variables are not randomly assigned, which makes identifying the causal relationship hard. My work addresses this by using instrumental variables and difference-in-differences designs to produce credible causal estimates from observational data. Across my research projects, causal inference methods allow me to answer questions about air pollution, smoking, early childhood outcomes, and reproductive policy. The common thread is a focus on populations who bear the largest costs of these exposures, and a commitment to getting the identification right so that findings hold valuable information for policymakers.

My job market paper, "Air Pollution and Cardiovascular Mortality: Evidence from Thermal Inversions in the U.S.," estimates the causal effect of fine particulate matter (PM_{2.5}) on cardiovascular mortality. Even though the health risks of air pollution are widely accepted, most existing studies using thermal inversions as an instrument rely on a single method of constructing the IV without testing whether that choice affects the validity of the instrument. My paper makes two contributions. First, it systematically evaluates how thermal inversion construction choices affect instrument validity and the reliability of causal estimates. Results show that decisions such as pressure level selection, defining inversion measures, and methods of aggregation to the county level all affect instrument strength and the plausibility of exclusion restrictions. My preferred specification is chosen based on these diagnostic criteria rather than picking the specification with the largest or most significant effect. Second, using this preferred construction, I find that a one unit increase in PM_{2.5} raises cardiovascular mortality rates by 3.8 percent. Results are robust to alternative pollutant measures, outcome transformations, and samples that exclude pandemic years, and placebo tests on unrelated outcomes pass. The effects are larger for Black, female, elderly, and low-income populations, which has direct implications for how the benefits of air quality regulation should be valued and distributed.

A second project, "Mental Health and Functional Consequences of Smoking: A Causal Approach Using Misreporting Corrections," examines the impacts of smoking on mental health and the ability to carry out daily activities. Two problems make this hard to study. First, smokers differ from nonsmokers in ways that affect health regardless of smoking, so standard regression estimates are biased. Second, self-reported smoking is not always accurate because people tend to underreport. Even after instrumenting, the key variable still contains measurement error that biases the estimates. I address both problems by combining a lagged cigarette tax instrument with a bivariate probit two-step model that estimates the probability of being a true smoker separately from the probability of reporting honestly. The corrected estimates show that smoking raises the probability of poor mental health and functional limitations by more than standard estimates

suggest, and the gap illustrates how much measurement error in health behavior data can distort causal inference.

A third project, "Air Pollution and Early Childhood Outcomes: Evidence from Georgia Childcare Providers" (with Garth Heutel), studies whether pollution keeps preschool-aged children out of childcare, a channel that has received little attention relative to birth outcomes or test scores. Using restricted administrative data from the Georgia Department of Early Care and Learning and thermal inversions as the instrument, we find that a one unit increase in county-week average particulate pollution decreases childcare attendance by about two percentage points. Effects are larger for racial minorities and children with working parents, groups where attendance disruptions carry higher costs for both child development and household income. We submitted grant applications to the American Lung Association, the Kaplan Foundation, and Georgia State University in support of this research and plan to submit the paper to NBER and a journal.

Outside my dissertation, I am developing a paper titled "The Economic Burdens of TRAP Restrictions: Effects on Housing and Childcare Costs." Targeted Regulation of Abortion Providers laws restrict abortion access through supply-side requirements on providers. Using a staggered difference-in-differences approach over 2010 to 2019, I estimate whether abortion restrictions impose broader financial costs on households beyond direct healthcare access.

Before starting my PhD, I published two papers on energy and CO₂ intensity measurement, one on the Iranian economy (Energy Policy, 2021) and one on the Netherlands manufacturing sector (Environment, Development and Sustainability, 2022). Both argued that standard monetary-based measures misrepresent true efficiency changes by ignoring physical activity data. Working closely with energy and emissions data made clear to me how pollution levels shape the lives of people near industrial and transport activity, which motivated my current focus on environmental health.

My future research extends this agenda into environmental health domains where medical literature has established associations but no credible causal estimates exist. Air pollution has received the most attention in economics, but many other exposures impose large and poorly measured costs on households, particularly those with fewer resources to avoid or recover from them. I am drawn to these gaps both because the identification challenges are interesting and because the populations most affected are often those whose outcomes matter most for policy. One project in this agenda examines how nitrate contamination in drinking water affects thyroid health. Nitrate is the most common chemical contaminant in US drinking water and the medical literature consistently links it to thyroid disease, but no economics paper has produced a causal estimate. Credible estimates here would let policymakers trace the full economic cost of water contamination in a way that the medical literature alone cannot.